



Faculty of Engineering & Technology

Department of Electrical & Computer Engineering.

COMP433, SOFTWARE ENGINEERING

Phase Three:

Requirements Analysis and Modelling

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Task 3.1: Actor Analysis:

Table 1: Task 3.1 Leadership and Contributions

Role	Name	Contribution
Lead	Shaymaa Dar Taha	Actor identification
Contributor	Abeer Hussein	Reviewing
Contributor	Roa Makhtoob	Discussing
Contributor	Marwa Faqeeh	validation

The following actors were identified from the business requirements and system requirements of the Bloom & Petals Flower Shop System. Each actor represents a role that interacts with the system to achieve specific goals and services.

Table 2: Actor Analysis

Actor	Semantics / Description
Customer	This actor represents a person who uses the system to browse flowers, search and filter products, place orders, customize floral arrangements, request delivery services, schedule future deliveries, subscribe to recurring flower deliveries, make payments, book consultations, register for workshops, manage account information, track deliveries, and submit product reviews.
Staff	This actor represents a shop employee who assists customers in-store, by phone, or through online support. The staff member can create customer orders, record customer information, manage subscriptions, assist with workshop registrations, arrange consultation bookings, update customer details, and help customers complete purchases and delivery requests.
Florist	This actor represents the flower specialist responsible for preparing and managing floral arrangements. The florist reviews and processes custom arrangement requests, creates floral designs according to customer preferences,

	provides floral design consultations, and conduct floral workshops offered by the business.
Delivery Staff	This actor represents the employee responsible for handling flower deliveries. The delivery staff can view assigned deliveries, manage same-day and scheduled deliveries, update delivery status, and confirm successful delivery of customer orders.
Manager	This actor represents the person responsible for managing the flower shop system and business operations. The manager can manage products, flower categories, prices, customer accounts, workshops, consultations, staff access privileges, reports, and other administrative functions required for business operation.
Payment System	This actor represents an external payment service responsible for processing electronic payments made by customers. It supports various payment methods, including credit cards, debit cards, and online payment gateways. The Payment System verifies payment transactions and returns the payment status (successful or unsuccessful) to the Bloom & Petals Flower Shop System.
Notification System	This actor represents an external notification service responsible for sending order confirmations, payment confirmations, subscription reminders, subscription expiration notifications, consultation confirmations, and workshop registration confirmations to customers through SMS messages.

Task 3.2- Use-Case Modelling:

Table 3: Task 3.2 Leadership and Contributions

Role	Name	Contribution
Lead	Roa Makhtoob	Use Case Diagram Design
Co-leader	Abeer Hussein	Use Case Analysis
Contributor	Shaymaa Dar Taha	Relationship Review (Include/Extend)
Contributor	Marwa Faqeeh	UML Compliance Check

3.2.1 Use Case Diagram

Marwa Faqeeh
Shaymaa Dar Taha
Abeer Hussein
Roa Makhtoub

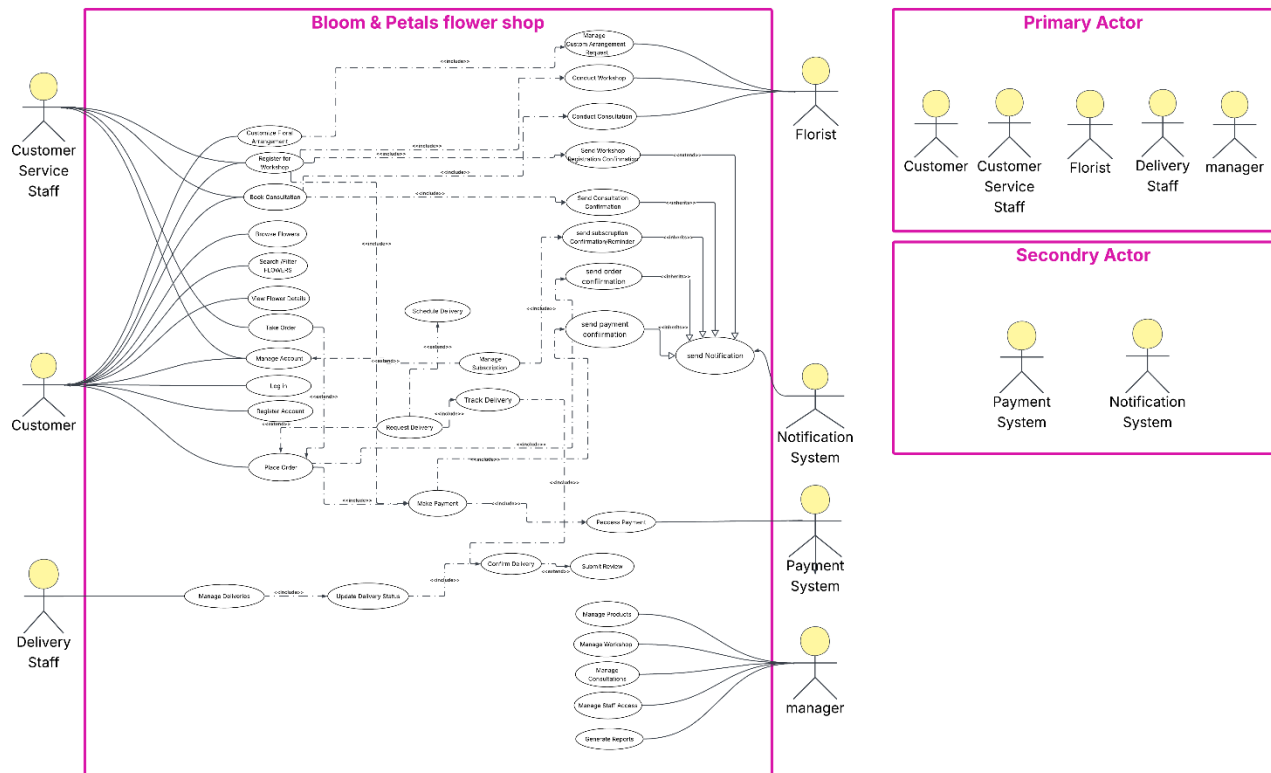


Figure 1: Use Case Diagram

3.2.2 Link:

https://lucid.app/lucidchart/c86756a7-1317-4fe5-b9b8-1f6ab43c94be/edit?viewport_loc=-6722%2C-5594%2C19012%2C6982%2C0_0&invitationId=inv_42418c4f-c1df-4ff0-85f7-bb104e764e4d

3.2.3 Use Case Diagram Description

The Use Case Diagram illustrates the interactions between the actors and the Bloom & Petals Flower Shop System. The primary actors include the Customer, Customer Service Staff, Florist, Delivery Staff, and Manager, while the Payment System and Notification System act as secondary actors that provide external services to the system.

The Customer is the main actor and can browse flowers, search and filter products, view flower details, place orders, request delivery services, track deliveries, manage subscriptions, customize floral arrangements, book consultations, register for workshops, manage account information, and submit reviews. The Customer Service Staff supports customers by creating orders, managing accounts, registering customers for workshops, booking consultations, and assisting with purchases.

The Place Order use case represents the central functionality of the system. It is connected to payment processing, delivery requests, order confirmations, and payment confirmations. Customers may also request delivery services, schedule future deliveries, and track delivery status after placing an order.

The Florist is responsible for handling custom arrangement requests, conducting consultations, and conducting workshops. When customers request customized floral arrangements, consultations, or workshop registrations, the system interacts with the Florist to process these services.

The Delivery Staff manages deliveries, updates delivery status, and confirms successful deliveries. The Manager is responsible for administrative functions such as managing products, workshops, consultations, staff access privileges, and generating reports.

The Payment System processes electronic payments, while the Notification System sends various notifications, including order confirmations, payment confirmations, consultation confirmations, workshop registration confirmations, and subscription reminders.

Overall, the diagram demonstrates how the different actors collaborate with the system to support ordering, customization, delivery management, subscription services, consultations, workshops, and business administration.

Task 3.3- Use-Case Specifications:

Table 4:Task 3.3 Use Cases Distribution.

Name	Use Case
Marwa Fageeh	Customize Floral Arrangement
Abeer Hussein	Book Consultation
Shaymaa Dar Taha	Manage Subscription
Roa Makhtoob	Place order

Table 5: Use Case Specification – Customize Floral Arrangement

System Name	Bloom & Petals Flower Shop System	
Use Case Title	Customize Floral Arrangement	
Description	A customer may request a customized floral arrangement based on occasion, preferred flowers, colors, style, and budget. The request can be made online by the customer or in-store with help from Customer Service Staff. The Florist reviews the request, checks availability, and confirms the arrangement or suggests modifications.	
Actors	Customer, Customer Service Staff, Florist, Notification System.	
Data	Occasion, flower preferences, colors, arrangement style, budget, setup request, florist feedback, request status	
Stimulus/Trigger	Customer selects or requests the Customize Floral Arrangement service	
Pre-conditions	1. The customization service is available. 2. The customer can access the system online or visit the flower shop. 3. A Florist is available to review the request.	
Workflow Sequence/Flow Events	OR of	1. Customer requests a customized floral arrangement online or in-store. 2. If the request is made in-store, Customer Service Staff assists the customer in creating the request. 3. Customer provides the occasion, preferred flowers, colors,

	arrangement style, budget, and any special requirements. 4. Customer submits the customization request. 5. The system records the request and forwards it to the Florist. 6. The Florist reviews the customization request. 7. The Florist checks flower availability and evaluates the feasibility of the requested arrangement. 8. If the requested arrangement can be prepared, the Florist confirms the arrangement. 9. Otherwise, the Florist suggests modifications or alternative flowers. 10. The customer reviews and approves the proposed arrangement. 11. The system updates the order status. 12. Notification System sends a customization confirmation to the customer.
Post-conditions/Response	1. The customization request is saved in the system. 2. The request status is updated as pending, confirmed, or requiring modification. 3. The customer receives confirmation or feedback regarding the request.
Comments	The customer must provide enough details for the Florist to prepare a suitable arrangement. The request may later be converted into an order after customer approval.

Table 6: Use Case Specification –Book Consultation

System Name		Bloom & Petals Flower Shop System
Use case Title		Book Consultation
Description		A customer may book a consultation session for floral design assistance through the online system. For in-store customers, Customer Service Staff may create and manage the consultation booking on the customer's behalf. The system displays available consultation time slots and allows the selection of a consultation type (In-Store or Virtual), date, and time. The system validates slot availability and stores the consultation details upon successful booking.
Actors		Customer, Customer Service Staff.
Data		Customer information, consultation booking details, consultation type, consultation date and time, booking status.
Stimulus/Trigger		Customer or Customer Service Staff requests to book a consultation session.
Pre-condition		1.Available consultation time slots exist. 2.Customer information is available. 3. The selected consultation date is valid. 4. A Florist is available to conduct the consultation.
Workflow sequence/flow events	Or of	1. Customer requests a consultation online, or Customer Service Staff creates a consultation booking for an in-store customer. 2.The system displays available consultation time slots. 3.The Customer or Customer Service Staff selects the consultation type (In-Store or Virtual).

	4.The Customer or Customer Service Staff selects the consultation date and time. 5.The system validates that the selected time slot is available. 6.If the slot is available, the system stores the consultation booking details. 7. The system assigns the consultation to the Florist's schedule. 8. The system confirms the booking. 9. Notification System sends a consultation confirmation to the customer. 10. The Florist conducts the consultation at the scheduled date and time.
Post-conditions/Response	1. Consultation booking details are stored in the system if successful. 2. The selected time slot becomes unavailable for other bookings. 3. Customer Service Staff and Florist can view the consultation booking. 4. A consultation confirmation is sent to the customer.
Comments	Customer Service Staff members must have appropriate permissions to view and manage consultation bookings. Consultation bookings are assigned to an available Florist.

Table 7: Use Case Specification –Manage Subscription.

System Name	Bloom & Petals Flower Shop System
Use Case Title	MANAGE SUBSCRIPTION
Description	A customer can create and manage a recurring flower delivery subscription. The customer selects the subscription frequency, duration, tier, delivery schedule, address, and flower preferences. For in-store customers, Customer Service Staff may create or manage subscriptions on the customer's behalf. The subscription can later be modified, paused, renewed, or cancelled.
Actors	Customer, Customer Service Staff, Notification System
Data	Customer account information, subscription frequency, start date, end date, subscription tier, delivery schedule, delivery address, flower preferences, subscription status
Stimulus/Trigger	The customer selects the " Manage Subscription " option from their account, or Customer Service Staff creates/manages a subscription for an in-store customer.
Pre-conditions	1. The customer has a registered account. 2. The customer is logged into the system, or Customer Service Staff is authorized to manage subscriptions. 3. Subscription services are available.
Workflow / Sequence of Events	1. Customer accesses the subscription management page, or Customer Service Staff opens the subscription management service for an in-store customer. 2. The system displays available subscription plans and options.

	3. Customer or Customer Service Staff selects a subscription frequency (weekly, monthly, or yearly). 4. Customer or Customer Service Staff selects a subscription tier (Basic or Premium). 5. Customer or Customer Service Staff enters the delivery schedule, address, and flower preferences. 6. The system validates the entered information. 7. Customer or Customer Service Staff confirms the subscription details. 8. The system stores the subscription information. 9. The system activates a new subscription or updates an existing subscription. 10. The Notification System sends a confirmation message to the customer.
Post-conditions/Response	1. The subscription is successfully created, updated, paused, renewed, or cancelled. 2. Subscription information is stored in the system. 3. Subscription status is updated. 4. The customer receives a confirmation notification.
Comments	Customers may modify, pause, renew, or cancel subscriptions at any time. Customer Service Staff members with appropriate permissions can manage subscriptions for in-store customers. The Notification System is responsible for sending subscription confirmations and reminders notifications.

Table 8: Use Case Specification –Place Order

System Name	Bloom & Petals Flower Shop System
Use Case Title	Place Order
Description	A customer may place an order for flowers or floral arrangements through the online flower shop system. For in-store purchases, Customer Service Staff may place the order on behalf of the customer. The order may include standard delivery, same-day delivery, scheduled delivery, or a previously approved custom floral arrangement. After successful payment, the system confirms the order, sends notifications, and forwards the flower preparation task to the Florist.
Actors	Customer, Customer Service Staff, Florist, Payment System, Notification System, Delivery Staff
Data	Customer information, flower information, order details, payment information, delivery information, delivery type, order status
Stimulus / Trigger	Customer decides to purchase flowers or a floral arrangement online or in-store.
Preconditions	1. Selected flowers or arrangements are available. 2. Customer has selected at least one product or an approved custom arrangement. 3. The system is operational. 4. Payment services are available.
Workflow / Sequence of Events	1. Customer browses flowers and views flower details online, or Customer Service Staff assists an in-store customer. 2. Customer or Customer Service Staff selects flowers, floral arrangements, or an approved custom arrangement. 3. Customer or Customer Service Staff places the order. 4. Customer may request delivery. 5. If delivery is requested:

5.1 Customer or Customer Service Staff enters delivery address, contact information, and delivery instructions.

5.2 The system validates that the delivery address is within the supported area.

5.3 Customer or Customer Service Staff selects the delivery type: Standard Delivery, Same-Day Delivery, or Scheduled Delivery.

5.4 If Same-Day Delivery is selected, the system adds the extra delivery fee to the order total.

5.5 If Scheduled Delivery is selected, Customer or Customer Service Staff selects a future delivery date and time.

5.6 The system validates that the selected delivery date and time are available.

6. The system calculates the total price of the order, including products, customization cost, and any delivery fees.

7. The system displays an order summary showing the selected items, delivery details, and total price.

8. Florist reviews and confirms the order details.

9. Customer selects a payment method.

10. If the order is placed online:

10.1 Customer pays using a card or supported online payment method.

10.2 Payment System processes the payment.

11. If the order is placed in-store:

11.1 Customer may pay using cash or card.

11.2 If card payment is selected, Payment System processes the payment.

11.3 If cash payment is selected, Customer Service Staff records the payment in the system.

12 If payment is successful:

12.1 The system records the payment.

12.2 The system updates the order status to Confirmed.

12.3 The system stores order and delivery details.

12.4 Notification System sends a payment confirmation to the customer.

12.5 Notification System sends an order confirmation to the customer.

		12.6 Florist prepares the flowers or floral arrangement according to the order specifications. 12.7 Florist updates the preparation status when the arrangement is ready. 12.8 If delivery was requested, Delivery Staff manages the delivery process. 12.9 Delivery Staff updates the delivery status until the order is delivered. 13. Else, if payment is not successful: 13.1 The system displays a payment failure message. 13.2 Customer may retry the payment. 13.3 Customer may select another payment method. 13.4 Customer may cancel the order.
Postconditions Response	/	1. The order is successfully recorded in the system. 2. Payment information is stored if payment is completed successfully. 3. Order and payment confirmations are sent to the customer. 4. The Florist receives and prepares the flower order. 5. Delivery information is stored if delivery is requested. 6. Delivery status is updated until the order is completed.
Comments		This is the central use case of the system because it connects product selection, custom arrangement requests, order placement, payment processing, florist preparation, notifications, and delivery management.

Task 3.4- Activity Modelling:

Table 9: Task 3.4 Leadership and Contributions

Role	Name	Contribution
Lead	Marwa Faqeeh	Activity Diagram Design
Co-leader	Abeer Hussein	Workflow Analysis
Contributor	Shaymaa Dar Taha	Scenario Development
Contributor	Roa Makhtoob	Model Verification & Finalization

3.4.1 Activity Diagram

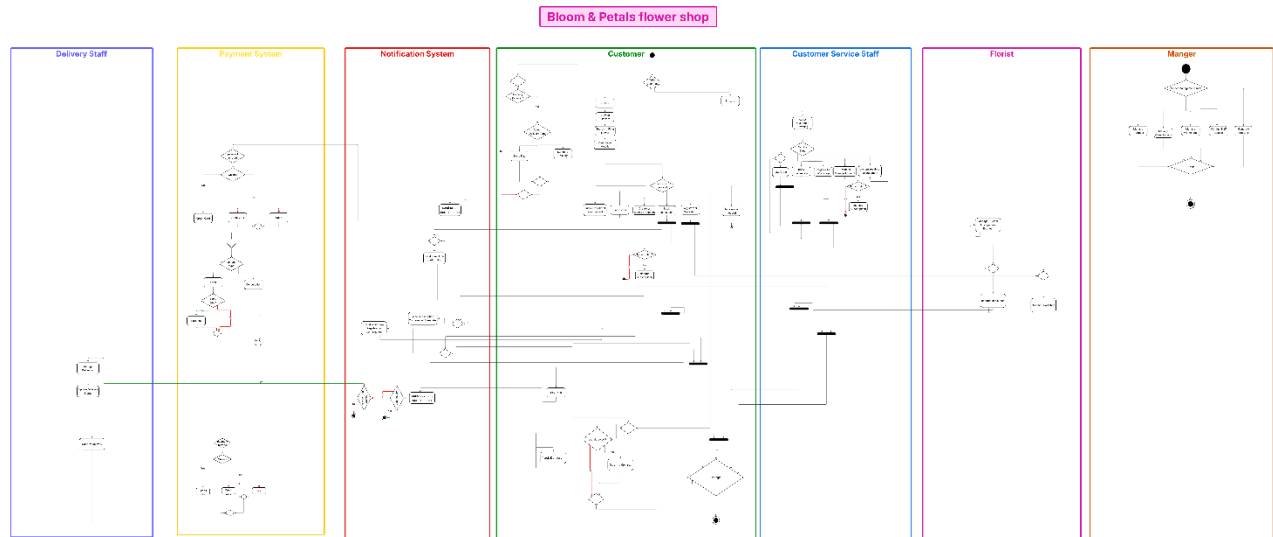


Figure 2: Activity Diagram.

3.4.2 Link

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3.4.3 Activity Diagram Description

The activity diagram shows how the different users and external systems interact with the Bloom & Petals Flower Shop System. The process starts when a customer accesses the system online or visits the flower shop and receives assistance from Customer Service Staff.

The customer can browse flowers, search for products, view flower details, place orders, request custom floral arrangements, book consultations, register for workshops, or manage subscriptions. If the customer decides to place an order, the system collects the order details, calculates the total price, and processes the payment using the Payment System.

For custom floral arrangements, the request is reviewed by the Florist, who checks flower availability and prepares the arrangement based on the customer's requirements. The Florist is also responsible for conducting consultations and workshops.

If delivery is requested, Delivery Staff manages the delivery process and updates the delivery status. The Notification System sends confirmations, reminders, and updates related to orders, payments, subscriptions, consultations, workshops, and deliveries.

The Manager is responsible for administrative tasks such as managing products, workshops, consultations, staff access, and reports. The process ends when the requested service is completed and the system updates all related information.